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Investigating context in sonic interaction design for sleep: Bridging users and designers through participatory and ideation workshops

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Abstract: Sleep is fundamental to health and well-being, yet sleep problems have risen sharply in recent years. These problems often arise from individual and contextual factors, some of which are modifiable through behaviour or design. Research examining sleep's contextual factors and how they can inform design interventions remains limited. This article presents two connected workshops exploring these aspects of sleep using participatory and speculative design methods. The first workshop invited participants to map contextual aspects of sleep through empathy maps and speculative exercises revealing three themes affecting sleep: safety, sleep routines, and stress or anxiety. The second focused on ideation asking design students to prototype sonic interventions addressing these themes. These prototypes were analysed by a professional sound designer to identify novel design elements. Findings highlight key considerations related to agency, ethics and technology, and offer insights and a flexible participatory methodology and framework for bridging user needs and design expertise.

Keywords: sleep, context, sound, sonic interaction design

1. Introduction

Sleep is essential for physical recovery, emotional balance, and cognitive functioning (Buysse, 2014; Ramar et al., 2021; Sano et al., 2016), but sleep problems are becoming increasingly common (Ahn et al., 2024; McArdle et al., 2020; Ohayon & Reynolds III, 2009). Recent research has shown that non-pharmacological approaches, such as listening to music and sound (Capezuti et al., 2022; Lai & Good, 2006; Riedy et al., 2021) and environmental modification (Caddick et al., 2018; Lai & Good, 2006) can positively influence sleep quality. Yet, much of this work focuses on auditory stimulation and physiological outcomes, overlooking how sleep is situated within broader contexts of everyday life. Difficulties in

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falling or staying asleep can arise from a combination of individual and contextual factors (Monsen & Edéll-Gustafsson, 2005). Some of these are inherent or hard to change, such as personality traits or age-related sleep patterns. Others are external and at times unpredictable, for example environmental noise or disturbances from cohabitants (Denis et al., 2018). However, factors such as aspects of one's sleeping environment, pre-sleep routines, and bedtime behaviours can be modified through design (Dobing et al., 2016). Exploring how these modifiable, contextual aspects shape sleep offers an opportunity to create more supportive and personalised solutions.

Research on the contextual aspects of sleep and their implications for design remains limited (Chung et al., 2025). Most sleep-related interventions focus on physiological monitoring without fully considering the user's lived situation (Grandner & Fernandez, 2021). Yet, the sleep environments and habits of people differ greatly: some live alone, others with partners, children, or pets; some follow strict schedules, while others work irregular hours or night shifts. Age, sex, lifestyle, and sensory sensitivities also affect how people experience sound and sleep (Mander et al., 2017; McGlinchey, 2015; Velluti, 1997). To develop more personal, context-sensitive, adaptive, and inclusive designs, there is a need to develop a rich understanding of user needs and sensitivities.

User-centered (Abras et al., 2004) and participatory design (Schuler & Namioka, 1993) methods offer ways to explore users' needs and their contexts and connect them to the design process. By actively involving people in reflecting on their sleep experiences, such approaches can reveal contextual influences often missed in clinical or controlled studies. Furthermore, when combined with speculative design techniques, novel ideas that address both the practical and psychological aspects of sleep can be generated.

This study, through two interconnected workshops and an interview with a professional sound designer, brings together the expertise of users, interaction designers, sound designers and researchers to reveal the rich entanglements between context, the individual and sleep, to develop novel design ideas, and encourage critical considerations about ethics and sustainability at the intersection of sleep and technology.

2. Background

2.1 Sleep and its contextual nature

Healthy and high-quality sleep can be described through several dimensions: satisfaction with sleep, daytime alertness, regular sleep patterns, sufficient duration, and ease of initiating and resuming sleep (Buysse, 2014). Contextual and environmental conditions such as noise, light, temperature, routines, safety, and co-sleeping arrangements (Billings et al., 2020; Caddick et al., 2018; Irish et al., 2015) influence these factors. Additionally, sleep does not occur in isolation, but is embedded within broader ecological and social contexts, interacting with environmental conditions, daily routines, and interpersonal dynamics (Gordon & Chen, 2014; Zisberg et al., 2010).

Some of these factors are difficult to change or external to a person's control while others, such as light exposure, temperature, and pre-sleep routines, can be controlled to an extent.

Research has shown that modifying relevant environmental and behavioural factors can improve sleep; for example, appropriate light exposure can support circadian rhythms (Tähkämö et al., 2019), temperature regulation can enhance comfort (Xiong et al., 2020) and consistent routines (Mindell & Williamson, 2018; Zisberg et al., 2010) can stabilise sleep onset and duration. Creating an overall sense of safety and comfort in the sleeping environment has also been shown to alleviate some sleep-related problems (Dobing et al., 2016). Overall, understanding the ecological and contextual nature of sleep becomes essential for identifying which aspects of the environment can be meaningfully shaped through design.

2.2 Sound and sleep

Sound can both disrupt and facilitate sleep, depending on its acoustic properties, timing, and personal interpretation (Capezuti et al., 2022; Monsen & Edéll-Gustafsson, 2005). Environmental noise, e.g. traffic, neighbours, or digital devices, has been widely reported as a source of disturbance, leading to difficulties falling asleep, fragmented sleep, and physiological arousal (Halperin, 2014) . On the other hand, some auditory stimuli, such as natural soundscapes (ocean waves, rain, forest sounds) or broadband noise (in commercial devices this might be white, pink or brown noise, sometimes mixed with everyday broadband sounds such as fan noise or rain), can promote sleep onset and continuity by masking unpredictable sounds, reducing perceived loudness contrasts, and eliciting feelings of safety and relaxation (Capezuti et al., 2022; Riedy et al., 2021). However, these effects are not universal, and the same sound may be experienced as calming or intrusive depending on context, habituation, and expectation (Capezuti et al., 2022; Riedy et al., 2021). Individual preferences, personal associations, and socio-spatial conditions, such as urban context or co-sleeping arrangements, strongly influence how sound is perceived at night. In recent years, interactive and adaptive sonic interventions for sleep have started to be developed thanks to the increase of consumer sleep monitoring technologies (Ko et al., 2015; Kolla et al., 2016; Hallak et al., 2025a; Hallak et al., 2025b). However, the variety of factors influencing sleep require contextual and experiential perspectives that go beyond measuring sleep efficiency to understand how sound influences sleep and the sleeping environment (Shin et al., 2022).

Sonic Interaction Design (SID) considers sound not merely as a soundscape to be listened to, but as a primary interactive design material to convey information, support behaviour, and shape affective experience (Rocchesso et al., 2008). Through SID designers can explore how sound might support sleep through interactive and adaptive systems that respond to bodily or environmental states. Recent examples of SID for sleep include Dozer (Semertzidis et al., 2023) that employs EEG-based feedback in a cap to deliver pink or white noise during sleep, and the Sonic Blanket that modulates sound in response to users' movements via embedded speakers (Choubey et al., 2024). Additionally, social robots such as Paro (*PARO Therapeutic Robot*, n.d.) or design concepts like the Yūgen Cloudstone Echo (Guo & Zhang, 2024) combine sound and light to support sleep and relaxation.

2.3 Bridging expertise

Participatory Design (PD) seeks to actively involve end users as full contributors in the design process (Schuler & Namioka, 1993; Bjerknes et al., 1987), embracing the "many-voiced nature of design" (Bødker & Buur, 2002). PD approaches have been successfully employed in health and well-being research, for example, in developing sleep, nutrition, and physical activity interventions with young people and shift workers (Ley-Flores et al., 2024; Shriane et al., 2024; Thabrew et al., 2018). These works show that designing with contextual sensitivity and involving participants meaningfully can lead to more acceptable and effective interventions.

Integrating diverse forms of expertise, needs, and viewpoints is a complex endeavor that demands the creation of effective *hybrid* spaces, i.e. environments that foster mutual learning and the reciprocal validation of differing perspectives (Muller & Druin, 2010).

Participatory spaces can be viewed as existing along a continuum: from the *abstract*, where end users must step into the designer's or researcher's domain to engage; to the *concrete*, where designers or researchers immerse themselves in the world of end users (Kensing & Munk-Madsen, 1993). Situated between these two poles are boundary regions that offer a hybrid experience: "a third space that contains an unpredictable and changing combination of attributes of each of the two bordering spaces" (Muller & Druin, 2010). Within this hybrid, third space, established assumptions can be questioned, challenged, reinterpreted, and refuted, enabling richer knowledge exchange through ongoing processes of reinterpretation and renegotiation (Bachmann-Medick, 1996).

A hybrid space can be defined by several factors. The site where the activity takes place, the nature of the activity itself (for instance, workshops often serve as productive venues for egalitarian exchanges of ideas and knowledge), the modes of communication employed between individuals with differing expertise (such as stories versus dramatizations), and the types of constructions that emerge from these activities (for example, low-tech or high-tech prototypes versus reflective writings or descriptions) (Muller & Druin, 2010). Some of the artifacts used to mediate interactions among diverse groups can be regarded as *boundary objects*, i.e. "objects that are both adaptable to different viewpoints and robust enough to maintain identity across them" (Star & Griesemer, 1989). First introduced by Star and Griesemer in 1989, a boundary object can be understood differently by different communities, while offering enough shared structure to support collective work. Such objects may be material or conceptual, ranging from maps, images, datasets or prototypes (Leigh Star, 2010; Star & Griesemer, 1989).

Hybrid spaces can also be virtual (Maher et al., 2000). For example, when parts of a design process is conducted asynchronously by different groups, at different times, in different sites, the objects and communications that make the collaborative work possible constitute elements of this virtual hybrid space, which someone, here the researchers, need to manage. In sonic interactions design such an approach was used in (Pauletto & Seznec, 2024) where groups with different expertise were able to ideate novel SID concepts for sustainability.

In this work, we utilised PD methods, including a virtual hybrid space and boundary objects, to bridge the expertise of users, interaction designers, sound designers and researchers and develop context-aware SID concepts for sleep.

3. Workshops: Overview and aims

Two interconnected workshops (WS1 and WS2) explored contextual factors of sleep and generated ideas for sonic sleep interventions. The resulting design concepts were later analysed and discussed with a professional sound designer (PrSD). WS1 focused on scenarios and design fictions; WS2 on ideating and lo-fi prototyping (see study structure in Figure 1).

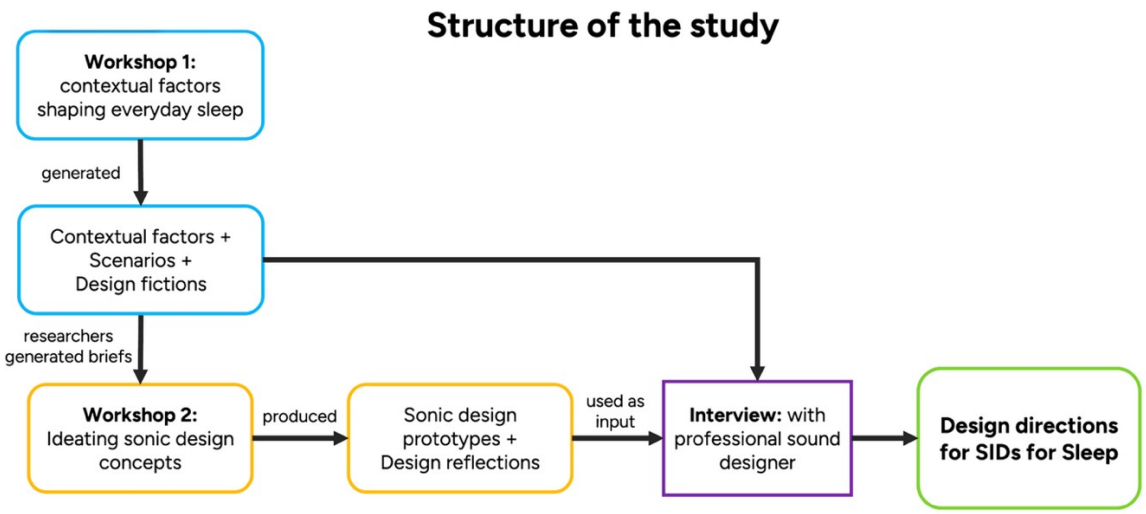


Figure 1 Overall structure of study.

3.1 Workshop 1 (WS1): contextual aspects of sleep

3.1.1 Structure and participants

WS1, lasting 3,5 hours, took place in June 2025 in a university classroom, and was divided into three parts: (P1) identifying contextual aspects of sleep, (P2) creating scenarios, and (P3) developing future design fictions. Seven participants (5M, 2F, average age: 33, SD: 7.5) took part: students, university staff, freelancers, and full-time employees representing varied living situations: alone, with partners, families, or in shared housing. Participants worked in groups of 2-3 people throughout the workshop.

In P1, participants reflected on factors influencing sleep, mapped them along two axes, *positive* vs. *negative* effect, and *common* vs. *unusual* (see Figure 2), and discussed their reasoning. This generated an overview of how participants perceived their sleeping conditions, combining personal experience and shared reflection. Using these insights, in P2, participants developed user characteristics and scenarios across three phases of sleep: going to sleep, being asleep, and waking up. Each group was assigned two of these phases and created one scenario for each, covering diverse demographics and living contexts. Participants used *empathy maps*, considering what the user might *say, think, feel, do*, (Ferreira et al., 2015) to outline users' experiences (Siricharoen, 2020). In P3, participants

created design fictions (Bleecker, 2022; Bresin et al., 2020). They imagined living in 2125 and sketched the front page of a newspaper focusing on sleep technology (see Prompts in Table 1) with headlines, short texts, images, and ideas for accompanying sounds. They were encouraged to think about the future of sleep technology beyond practical constraints considering both drawbacks and opportunities.

3.1.2 Results

In groups of 2-3, participants discussed and wrote down in post-its relevant contextual factors that then were positioned along the axes positive, negative, unusual, common. They then presented their reasoning. Presentations were recorded, transcribed and analysed together with the post-its through a manual thematic analysis (Clarke & Braun, 2017) conducted by two researchers. Observations from P1 mostly clustered in the *common* and

Table 1 Prompts for P3: Design Fictions on Sleep and Sound.

No.	**Prompt Question**
1	What has already been done in the area of sound and sleep today (in 2125) that causes headlines?
2	What was not done in the past that caused a scandal? (i.e. think about what is not done now that people in the future think should have been done, i.e. which caused the current situation.)
3	What do you want to see done in the near future? (i.e. what is missing in the present that could create impactful headlines in the future.)

negative quadrants (see Figure 2) and related to three domains: the sleeper, other living creatures, and the environment. Factors such as late eating, caffeine, and screen use were described as disruptive, while exercise, structured wind-down routines, and comfortable environments were considered beneficial. Disturbances from partners, children, or pets were common, as were noise and light pollution. Predictable background sounds were perceived as sometimes comforting, and sleep tracking was mentioned as an unusual but positive practice for sleep. Living in small or shared spaces without dedicated sleeping areas was noted as negative. Sunlight and weather were mentioned to influence sleep depending on timing and season.

In P2, participants created scenarios and user characteristics using empathy maps (see example in Figure 3). Groups discussions and presentations were recorded, transcribed and analysed. Group1 described a teenager losing sleep due to gaming and social media, and a father disturbed by childcare and his partner's irregular hours. Group2 portrayed older men with fragmented sleep, one relying on a sleep apnea machine and another falling asleep in front of the TV. Group3 imagined two sisters sharing a bed for comfort and a mother balancing work, family, and pet-related disturbances. These translated general factors from P2 into concrete scenarios for later design ideation.



Figure 2 Contextual factors of sleep mapped by participants along positive/negative and common/unusual axes.

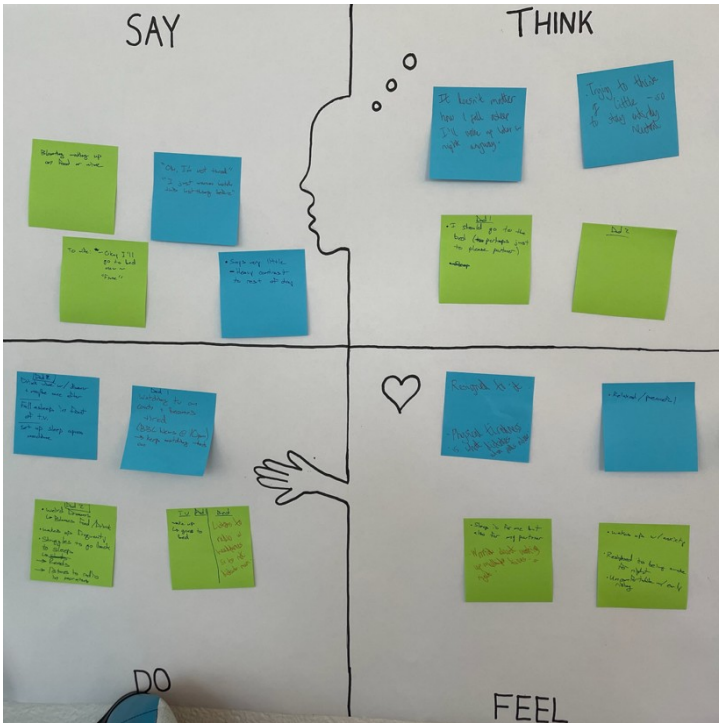


Figure 3 Example empathy map created by participants in P2.

In P3, design fictions were created (see Figure 4). As before, groups discussions and presentations were recorded, transcribed and analysed. Group1 focused on the "going to sleep" phase and imagined *Noiseless*, a sound beacon projecting adaptive noise-cancelling waves for quiet zones. Group2 addressed "waking up" through *Lucy*, a lucid-dream device designed to reduce morning stress via sound frequencies that modulate cortisol, and discussed concerns about affordability and access of sleep technology. Group3 explored "sleeping during heatwaves", imagining a sonic device that alters dream content to simulate



Figure 4 Design fictions from P3 illustrating sonic concepts for sleep: adaptive noise-cancelling (Group 1), lucid-dream-based waking support (Group 2), and heatwave-related dream modulation (Group 3).

coolness. This idea prompted ethical questions around data privacy and the potential of sleep technology to be used for manipulation. Across groups, sound emerged as a tool for sleep control, comfort, and imagination, but also as a potential source of ethical tension around noise, privacy, sustainability, affordability, and agency, aspects which were further considered in WS2.

3.2 Workshop 2 (WS2): Ideating sonic interventions for sleep

3.2.1 Structure and participants

WS2 took place in October 2025 in a university design studio. It lasted three hours and was divided in two parts (PP1 and PP2). Thirteen master's students (4M, 9F, average age: 25, SD: 3.5) in media and interaction design participated, working in four groups. All had some background in sound or music technology. Participants were able to utilise insights from users through scenarios, briefs and prompts based on WS1 and devised by the researchers. These short, precise but also open, texts and stories constitute the initial "boundary objects", adaptable to different viewpoints but with identity, that were used to facilitate collaboration between users, researchers, and interaction designers.

PP1 centered on ideation and prototyping of SID for sleep based on a scenario and brief, while PP2 invited participants to critically re-imagine their concepts based on the ethical considerations that emerged in the last phase of WS1.

Scenarios from WS1 were used to develop four design briefs around safety, sleep routines, and stress or anxiety before sleep (see 7.1-7.4 Appendices). Each group brainstormed sonic interventions for one of the scenarios. An ideation canvas (see 7.5 Appendix), developed by the researchers and inspired by the Data Sonification Canvas (Lenzi & Ciuccarelli, 2024) and the prototyping canvas (Lauff et al., 2019), was used to structure the design process. This

helped articulate key aspects such as problem context, intended interaction, sonic elements, and user experience. Low-fi prototypes or visual representations of the concepts were created and presented.

In PP2, participants critically reflected on and re-visited their designs through ethical concerns from WS1 (see prompts in Table 2)

Table 2 Ethical and Practical Reflection Prompts

Factor	**Prompt**
Noise	Consider the safety of cancelling external noise. Consider whether adding to the existing soundscape can create issues for others
Privacy	Consider whether there are unintended unethical implications in your idea. For example, could it be used for surveillance, or to manipulate the user in some way?
Sustainability	Consider the environmental implications of your device. For example, does it require excessive energy consumption or use harmful materials?
Affordability	Consider the cost of such intervention and reflect on who can afford it.
Agency	Consider whether the intervention is adjustable by the user along their preferences. Consider who has control over the user's data. Consider if the user's agency is supported.

3.2.2 Results

All groups discussions and presentations were recorded, transcribed and analysed.

Design1 (D1). Group1 reflected that John (62) (see Appendix 7.1 for scenario and brief), struggled primarily with mental exhaustion and by missing his natural sleep window due to his problematic pre-sleep routine. They proposed a system combining wearable devices, a mobile app, a speaker and adaptive soundscapes (see Figure 6 and 7). The wearables would provide gentle sound cues and motivational music during the day to encourage activity, while the app would issue notifications, based on his activity levels. At night, John could select personalised relaxation soundscapes, with an additional noise cancelling option. John's wife, who shares the bed, would also have control over the sound playback. The group distinguished between active and passive listening, with daytime cues being noticeable and in-bed sounds unintrusive.

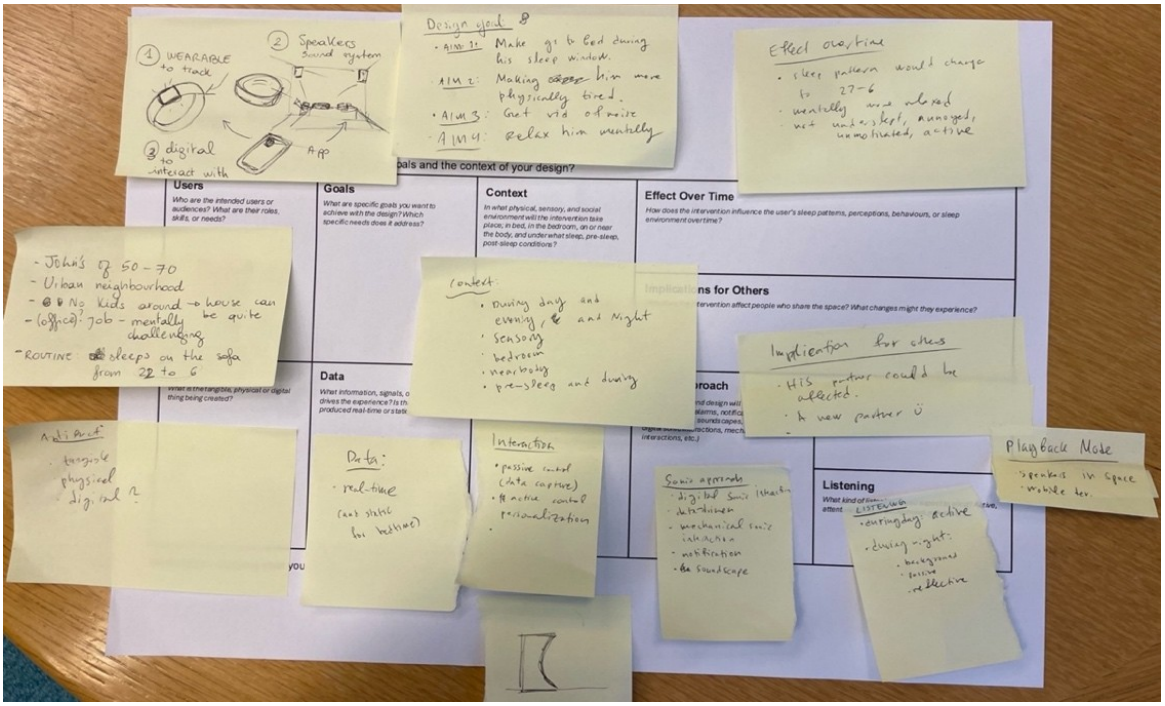


Figure 6 Design Ideation Canvas Group 1.



Figure 7 Complete prototype with all the elements (wearables, bed with users and silver speaker) (Group 1).

During reflection, the group proposed using locally recorded environmental sounds for sleep (excluding voices for privacy) instead of pre-made, third-party material, while preserving essential sounds like notifications. The system would not store data without consent. Overall, the group acknowledged this system would likely be costly and technically complex. They also noted the importance of sound literacy, suggesting that if John lacked familiarity with sound design, the system would need to offer a simpler interface with basic, easy-to-

understand options and sounds. In relation to agency, the group considered that a partner would also be affected by such system and therefore control should be shared. They considered that this could have both positive (a closer relationship) and negative (conflicts) implications.

Design2 (D2). Group2 designed a "napping sofa" for Leah (17) (see Appendix 7.2) as a safe space to ease the transition into sleep (see Figures 8 and 9). They observed that pressure around sleep time and duration can itself prevent sleep, and relaxation often occurs more naturally in neutral spaces such as couches. The sofa would contain different "texture" materials, such as sand, water, or pebbles that would shift with user movement, producing soft, organic sounds. Additionally, a device would collect physiological data and use it to further control the sofa. Around the sofa, light and scent could enhance comfort, while familiar objects and a phone storage compartment would encourage detachment from screens. Other, simple activities (reading a book or interacting with tactile objects) would be encouraged. D2 aimed to create calmness and safety and was not meant to replace the bed. During reflection, the group raised concerns around misuse of personal data and suggested removing technological components entirely, imagining a purely mechanical version of the concept. Sound elements could be modular and personalised, and the sofa or armchair would naturally quieten as the user relaxed.

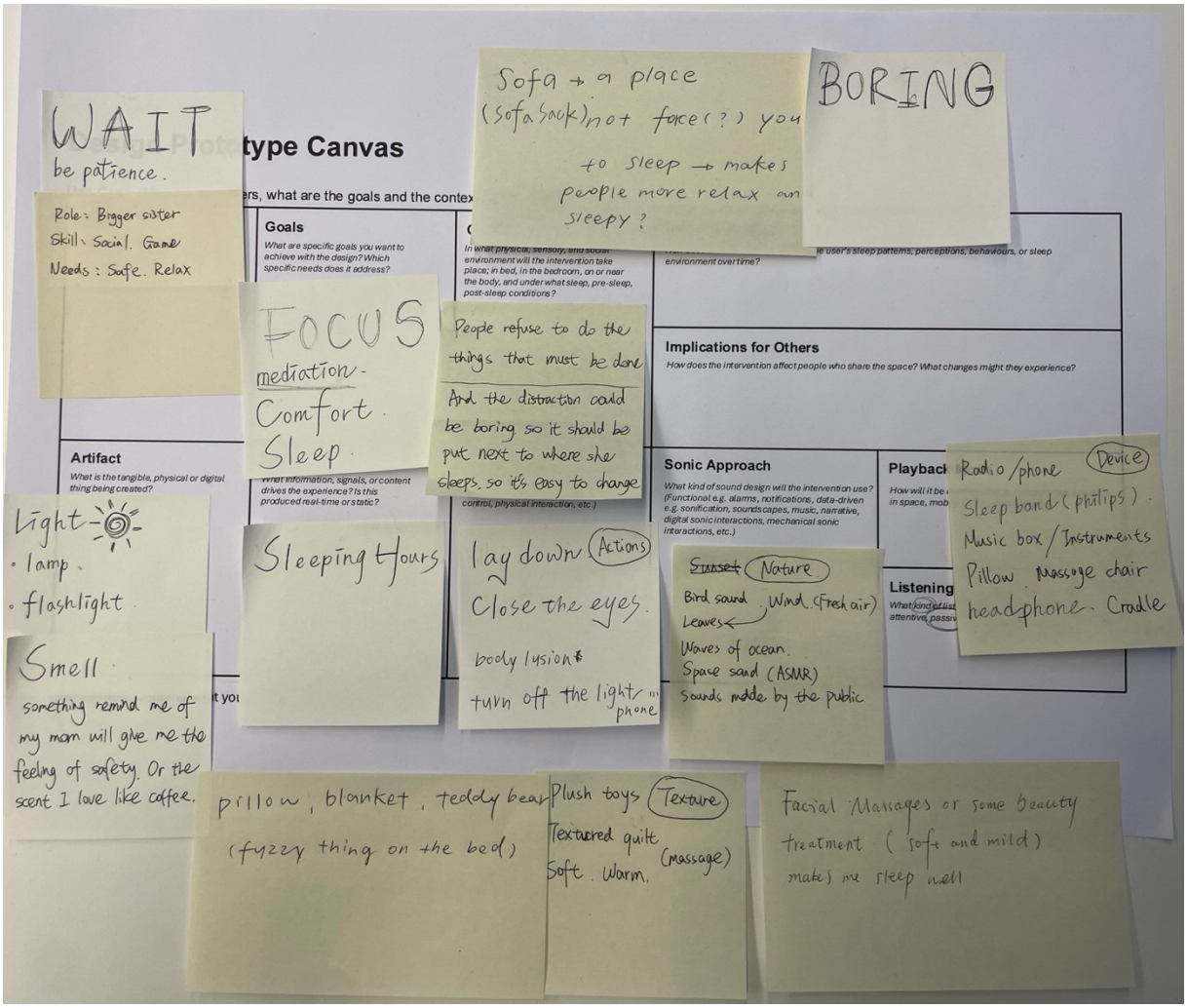


Figure 8 Design Ideation Canvas Group 2.

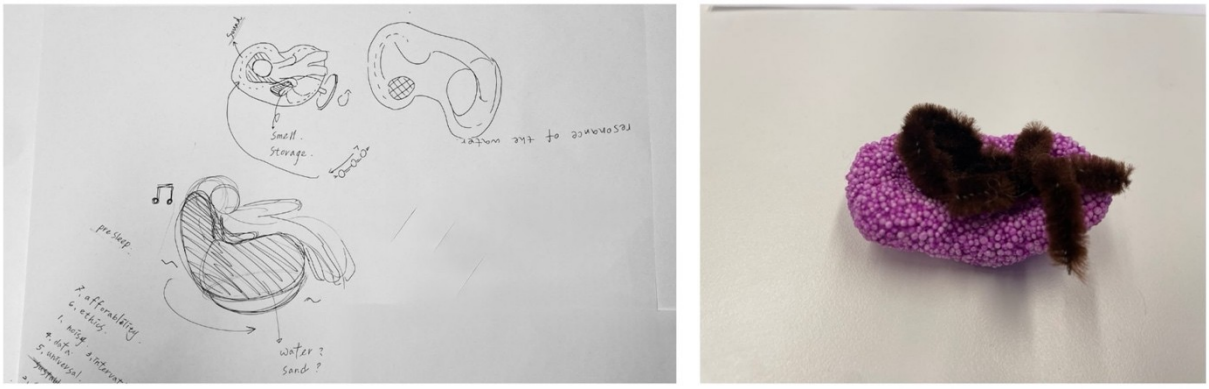


Figure 9 Complete prototype with the design sketch and low tech implementation of the design (Group 2).

Design3 (D3). This group addressed stress and overthinking. They described Mayank (42) (see Appendix 7.3) as physically tired but mentally overactive, struggling to unwind after managing household responsibilities. They developed three related ideas. The first concept proposed a bedside device (see Figures 10 and 11) linked to a health app that would record

environmental sounds and detect sleep onset over several weeks. An AI system would then generate sleep-conducive sounds, preserving familiar features while modulating frequencies for optimal sleep. The second and third variations would allow users to record or select soothing sounds, or to listen to live streams of relaxing natural soundscapes from other locations. Playback would be designed to be unobtrusive and to establish associative cues linking specific sounds to sleep.

During reflection, the group decided that the AI should operate locally and users would not directly control how sounds were generated, but could adjust playback to protect privacy and minimise environmental impact. The system was considered moderately affordable for most users in relatively wealthy countries, but not universally.

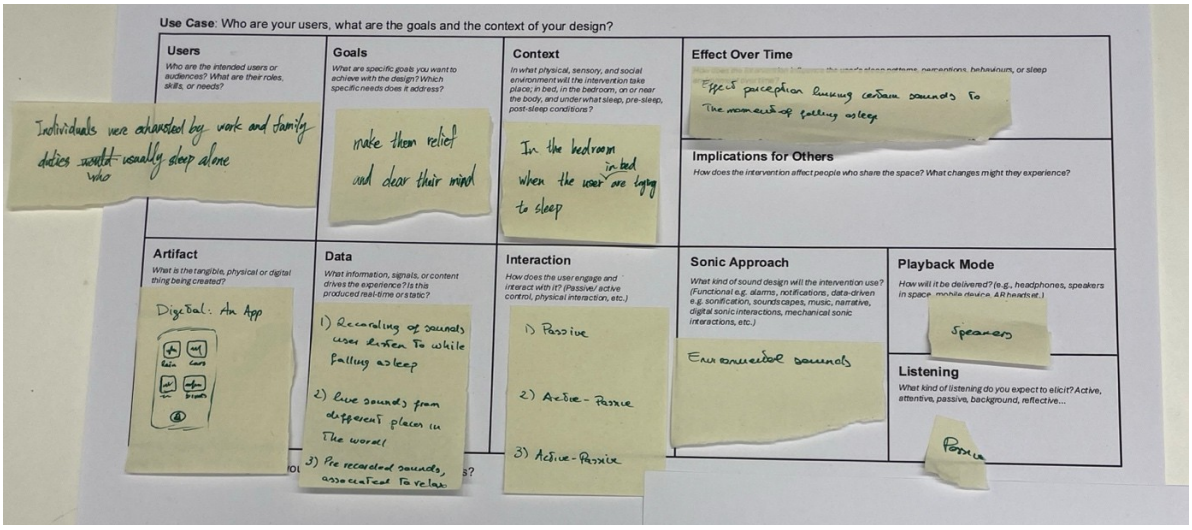


Figure 10 Design Ideation Canvas Group 3

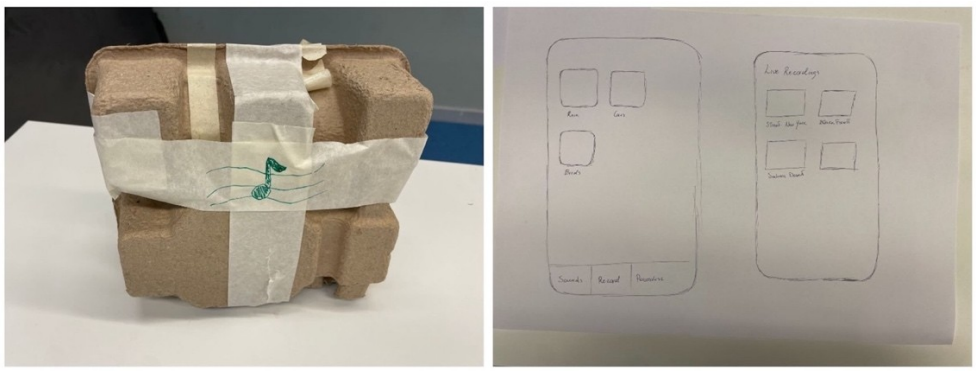


Figure 11 Complete prototype with the design sketch and low-tech implementation of the design (Group 3).

Design4 (D4). This group focused on reducing disturbance and promoting comfort in a shared environment. Amy (48) (see Appendix 7.4) lives alone, in a noisy city apartment, with an irregular schedule, and a restless cat. Their main idea involved wall-mounted speakers emitting gentle broadband pink or brown noise, subtly varied to avoid monotony, to mask intrusive sounds (see Figures 12 and 13). A variation integrated miniature speakers into a

pillow or blanket with optional heating for relaxation. Spatial zones, such as carpets for sound absorption and defined “warm zones” (in orange) for the cat away from the bed (such as a cat tower), further reduced disruptions.

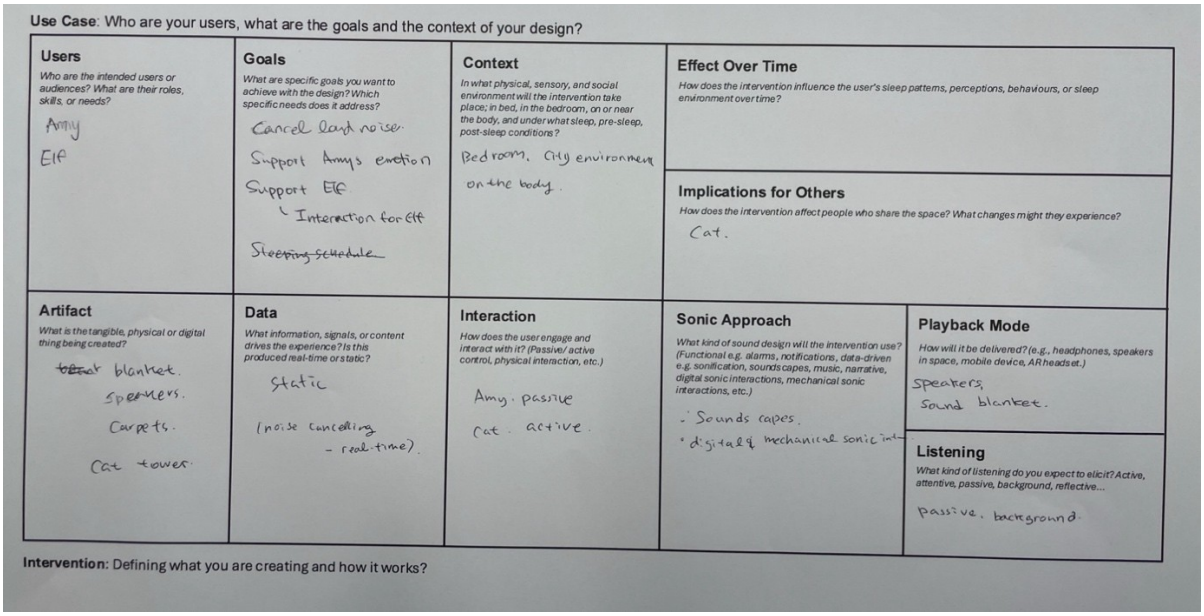


Figure 12 Design Ideation Canvas (Group 4)

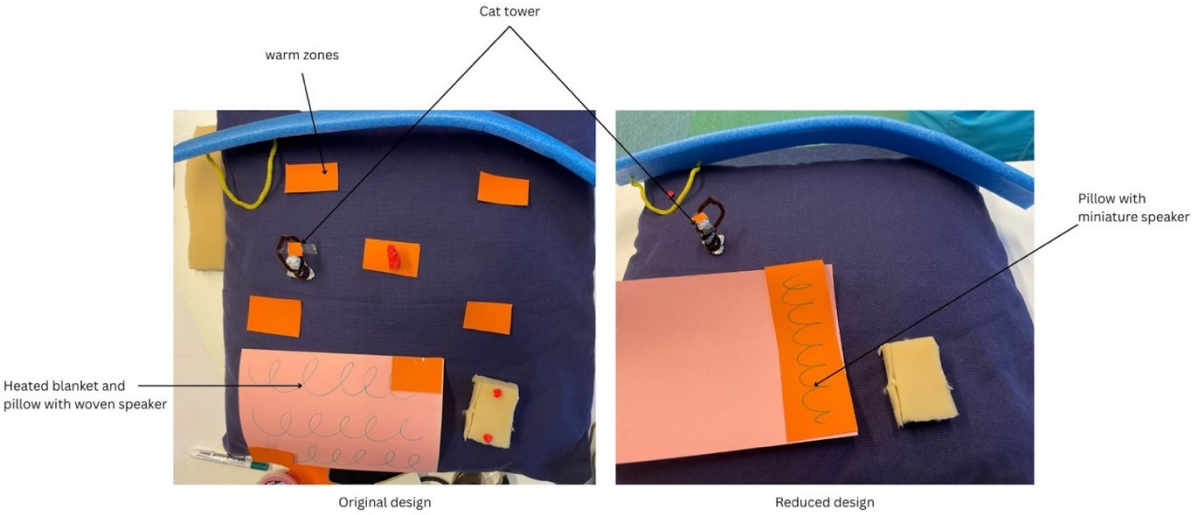


Figure 13 Initial prototypes and revised prototype with reduced warm zones (Group 4).

During reflection, the group favoured passive over active noise control, allowing critical sounds, like alarms, to remain audible while filtering irrelevant noise. No data collection would occur, and design elements were reduced in number for simplicity and sustainability. The intervention should consider the cat’s presence, and the ethical aspects and effect on agency related to it, including its behaviour, sleeping patterns, and listening needs.

3.3 Sound Designer Analysis

To further elaborate on these design ideas, particularly on the sonic aspects, we asked a professional sound designer (PrSD) to analyse and reflect on these concepts. PrSD is a composer and sound designer with extensive experience collaborating with international companies and research institutions on the development of sounds for electric vehicles and other products. Additionally, their portfolio encompasses a substantial body of artistic research focused on themes such as insomnia, noise reduction, unobtrusive auditory signals, calm and peaceful alarms including wake up alarms, and everyday soundscapes, making their perspective particularly relevant to the design of sleep related sonic interventions.

PrSD received the scenarios, the design brief, images of the prototypes, along with three guiding questions:

1. Which aspects of the final design ideas are most interesting or novel?
2. Which aspects are less convincing or could be refined?
3. How would you approach designing the still-undeveloped sonic elements (e.g. sounds, processing, listening conditions)?

These were the “boundary objects” used to facilitate collaborative work between junior interaction designers, the PrSD, and the researchers. The researchers and the PrSD met in zoom and the discussion was recorded, transcribed and summarised.
Before discussing individual scenarios, PrSD shared two main general observations: the designs tackle (1) issues that can be addressed through sound (e.g. reducing intrusive noise), and (2) deeper psychological issues, like rumination, that would benefit from a close collaboration with sleep therapists (something we plan for the future). Additionally, PrSD noted that, while considering privacy and ethical issues is essential in design, avoiding data-driven approaches altogether would be an excessively restrictive solution. Instead, providing clear privacy protection measures should be considered.

PrSD found D1 to be a rich and comprehensive design, combining several complex components. Day-time notifications could have different degrees of intrusiveness depending on their function and occurrence during the day: more energetic in the morning, softer and slower in the evening to gently support physical activity and sleep. Additionally, soundscapes could align with breathing and biorhythms, perhaps requiring a data-driven approach to be effective. Overall, the distinction between active (during the day) and passive (during the night) listening was seen as very valuable. Finally, PrSD recognised that providing shared control of the soundscape would be a very interesting design challenge related to agency, not one he worked on before, and admitted that no simple solutions came to mind.

PrSD supported the idea of a transitional relaxing space (the napping sofa) in D2, noting that remaining in bed while unable to sleep can worsen insomnia. This highlighted that transitional spaces may not be inherently relaxing and can introduce pressures or stress if sleep does not occur as expected. PrSD raised concerns about perceptual incongruence, where sounds that do not align with the expected material qualities of an object (e.g. a sofa

producing granular textures such as sand or pebbles) may create discomfort by violating user expectations. They suggested that this effect might not be dissimilar to the "uncanny valley" effect (Mori et al., 2012), where human-like objects, e.g. dolls or robots, that appear almost human can produce a feeling of discomfort. Overall, they recommended creating smoothing sonic transitions to maintain a slow, lulling rhythm. Taken together, this refined our interpretation of the design by the need to balance relaxation with user expectations, and to ensure coherence between the physical object, its intended use, and its sonic behaviour.

D3 and D4 were discussed together, due to their common emphasis on the use of sound to mask external disturbances. Instead of using continuous sound, PrSD proposed introducing additional sounds in the soundscape to soften the transition between intrusive sounds and creating a more cohesive whole. For D3 (the bedside device), PrSD suggested adding a tactile component—such as a "sleep memory box" (something similar to (Kleinberger et al., 2019; *Music Memory Box – Studio Meineck*, n.d.)) that could trigger familiar feelings. For D4, PrSD emphasised the ethical dimension of co-sleeping with pets and the implications for agency, recommending learning about and considering the animal's comfort and auditory well-being.

4. Discussion

Taken as a whole, this ideation process moved from mapping lived experiences of sleep and considering ethical concerns (WS1), to ideating and prototyping design ideas (WS2) that were then refined and turned into actionable SID directions (PrSD analysis). This process required the joined effort of four distinct groups: users, interaction designers, sound designers and researchers. The asynchronous nature allowed different voices to be heard efficiently (no co-presence needed) and required the creation of a virtual hybrid space where researchers and boundary objects bridged expertise. Here we reflect on the main findings and the methodology employed.

4.1 An initial framework for context-aware SID for sleep

In WS1, we identified three separate categories of contextual factors related to: the sleeper, other living creatures, and the environment. This helped users think about different user situations. After analysis, three main reasons for disturbed sleep emerged: lack of safety, stress and anxiety, and disrupted (or lack of) sleep routines. By combining these, researchers were then able to distill concrete scenarios and briefs to address SID for sleep in context.

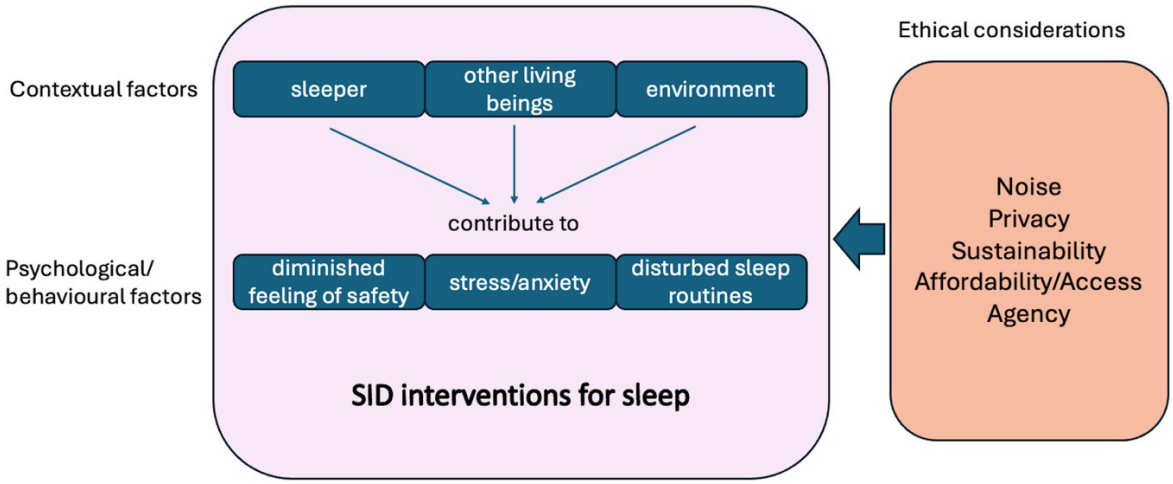


Figure 14 Initial Framework for Contextual-Aware SID for Sleep.

The speculative part of WS1 was effective in allowing users to freely express concerns about technologies that aim to affect humans during sleep, i.e. while not fully conscious. A thematic analysis (Clarke & Braun, 2017) of participants responses, brought up five distinct concerns-noise, privacy, sustainability, affordability/access, and agency, providing researchers with clear directions with which to probe designers' ideas.

Overall, results of WS1 can be summarised in an initial framework for context-aware SID for sleep (see Figure 14).

The concepts formulated in WS2, and refined through the PrSD analysis, reveal four key challenges that warrant consideration when applying SID to the context of sleep.

4.2 Designing for shared sonic control

We do not often sleep alone and this foregrounds questions of agency and control. D1 raised the important question of who controls sound in a shared sleeping area. Here a negotiation must occur between users, and the role of the designer is to facilitate it maintaining the social aspect of sleep, rather than separate each user with earplugs. Additionally, the presence of the pet in D4 allowed to reflect on other living creatures' behaviour, agency and well-being. Our approach proved useful in discussing these dynamics, as negotiation is a central component of participatory design.

4.3 Designing pre-sleep spaces

Designing for sleep does not solely entail designing for the sleep moment itself. D1 and D2 targeted users' daily activities (exercise, routines, or napping). In this space, listening can be active and sounds more intrusive encouraging the user towards helpful sleep behaviours.

4.4 Local sound / Familiar sound

Several designs (1–3) departed from using third-party sounds and instead focused on environmental, user-generated sounds, or sounds produced by the designed object. This enhanced familiarity with the sonic material and support more personalised experiences in which users could retain a sense of agency, while also reducing reliance on streaming or automated (AI-based) systems considered less sustainable, less affordable, and environmentally detrimental.

4.5 Multi-sensory and more-than-human sleep design

D2 and D4 were not solely sonic proposals but holistic, multisensory, and, in the case of D4, more-than-human designs. As sleep is influenced by multiple contextual factors, it often requires a multimodal intervention that can accommodate diverse sleep patterns (human and non-human) and varied multisensory sensitivities.

4.6 Reflection on the methodology

We used a PD approach to examine contextual factors influencing sleep and to explore how these can inform design ideation in SID for sleep. The dialogues—first within workshop groups and later between PrSD and researchers—helped shift the discussion from assumptions to reflections (Carrillo 2013). This shift supported the development of knowledge carried forward through stories, briefs, and prototype images functioning as boundary objects, with researchers acting as facilitators and intermediaries. In this process, stories and images became primary sources of inspiration for the sonic reflections (only a few sound sketches were produced, and these were not considered fully representative). The oversized silver speaker in D1 and the blue noise-cancelling barrier in D4, for instance, suggested sound-based interventions as forms of defence against unwanted noise and disturbances. Prompts served as catalysts for storytelling, analysis, and feedback, opening to dialogues and participants' experiences, and helping articulate design functions (Muller and Druin 2010). As sessions unfolded across different times and locations, we were able to construct a virtual, hybrid space similar to that described in (Pauletto & Seznec, 2024). The tools employed (empathy maps, design fictions, and a novel design canvas) effectively supported participants' reflections and progress, enabling rich outcomes efficiently. Overall, although our findings are not exhaustive, they provide concrete, actionable guidance for context-aware, sleep-related sonic interaction design and offer an adaptable methodological framework—with practical tools—to support further research in this domain.

5. Conclusion

This study examined how contextual factors shaping sleep can inform the design of sonic interventions. Across two workshops and an expert analysis, participants mapped everyday sleep experiences, identified challenges, and translated these into design concepts. Participatory and speculative methods facilitated nuanced understandings of sleep as a situated, multisensory phenomenon.

The concepts demonstrated the varied roles of sound—from masking disturbances and supporting routines to fostering comforting and multisensory atmospheres—and helped discuss ethical considerations. The methodology offers a flexible participatory approach. While the participant groups are small, limiting generalisation, the participatory design methods approach generated rich qualitative insights. These allowed researchers to offer novel design directions that can inform the development of future sonic interaction designs for sleep. Overall, the research advances discussion on how thoughtfully designed SIDs can actively shape sleep, well-being, and personal environments.

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7. Appendix

7.1 Scenario 1 – Routine and sleep timing

John, 62, lives with his wife in a mid-sized apartment in a noisy urban neighborhood. Their two children have moved out, and evenings at home feel quiet. He works regular 9 to 5 hours as a marketing manager, a mentally demanding role that leaves him exhausted but not physically tired.

Most nights, John unwinds by watching TV on the couch, often falling half-asleep mid-episode. The habit winds him down, but it also delays his bedtime, he stays up longer than intended, losing sleep time. By the time his wife nudges him to bed, his natural sleep window has passed. Once in bed, he lies awake, replaying work thoughts. In the bed, the city noises, neighbors talking, distant traffic, becomes more noticeable and makes him stay up even more. These interruptions, combined with his irregular routine, leave him restless and underslept. He wakes up sluggish, unmotivated, and easily irritable the next morning.

Design Brief:

Design an interactive sonic intervention that helps John establish and maintain a healthier sleep routine.

7.2 Scenario 2 – Safety

Leah, 17, lives with her parents and younger brother in a small family home. Her bedroom is her personal space, the only space she feels she can truly call her own. Yet even there, she doesn't always feel safe or comfortable enough to unwind. Late evenings often bring anxiety and comparison, amplified by what she sees on her phone.

Every night, Leah lies in bed scrolling through social media or gaming with friends. It starts as a way to relax and "catch up," but soon she's lost in numerous videos, messages, and social media content. The light from her screen keeps her alert, and the content, which is most times full of comparisons, expectations, and drama, makes her feel inadequate and restless. When she finally puts her phone away, her mind is still full of thoughts. The silence of her room emphasizes that, and she struggles to fall asleep, often replaying conversations or posts in her head.

Design Brief:

Design an interactive sonic intervention that helps Leah feel safe, grounded, and secure in her sleeping space.

7.3 Scenario 3 – Stress and pre-sleep anxiety

Mayank, 42, lives in a quiet countryside town with his wife and three children. His wife is a healthcare worker with unpredictable night shifts, which means he often manages the household and bedtime routines on his own. The family home is peaceful during the day, surrounded by nature, but evenings are chaotic; dinner prep, homework, and getting the kids to bed all fall on his shoulders. By the time the house finally quiets down, Mayank is physically exhausted but mentally overstimulated.

When he finally gets into bed, his mind doesn't slow down. He replays the day, the kids' arguments, unfinished work emails, bills, errands, and anticipates the next day's demands. His wife tries to help as much as she can, but her absence on certain nights amplifies his sense of responsibility and occasional loneliness. He sometimes tries to unwind by scrolling on his phone or checking messages, but this only adds to his restlessness. Even when he feels tired, his body resists sleep; his thoughts keep looping, and he experiences insomnia many times. The result is chronic fatigue and irritability that bleed into his days, feeding a cycle of stress and sleeplessness.

Design Brief:

Design an interactive sonic intervention that helps Mayank release stress and anxiety before sleep.

7.4 Scenario 4 – Shared space and environmental disturbances

Amy, 48, lives alone in a small apartment on the fifth floor of a busy city building. Her only companion is her cat, Elf, who curls up near her bed and keeps her company through the day. She works as a freelance graphic designer—flexible hours that blur the line between work and rest. Some nights stretch into late design revisions, others into early calls with clients abroad.

At bedtime she attempts to relax: tea, soft lighting, and quiet. But city sounds—sirens, elevators, footsteps, slip through the walls, keeping her half-awake and often heightening her loneliness. To cope, she leaves the TV or a podcast playing, or turns on nature sounds to cover the noise, though none ever feel quite right. Elf's sudden movements—jumping off the bed, scratching, or meowing for food at dawn—often wake her just as she begins to drift off.

While Elf brings comfort, it also brings restlessness. Amy worries about him, whether he's hungry, bored, or unwell, and that worry lingers at night. Her bedtime shifts between 10 p.m. and 2 a.m., depending on her mood or workload. When she finally lies down, her mind runs through unfinished designs, unread messages, and family concerns. Over time, she's noticed that she isn't sleeping deeply; she wakes easily and often feels unrefreshed in the morning.

Design brief:

Design an interactive sonic intervention that helps Amy feel safe, emotionally supported, and gently guided into a stable and comforting bedtime routine.

7.5 Sonic interaction design for sleep canvas

Sonic Interaction Design for Sleep Canvas

Use Case: Who are your users, what are the goals and the context of your design?

Users	**Goals**	**Context**	**Effect Over Time**	
Who are the intended users or audiences? What are their roles, skills, or needs?	*What are specific goals you want to achieve with the design? Which specific needs does it address?*	*In what physical, sensory, and social environment will the intervention take place; in bed, in the bedroom, on or near the body, and under what sleep, pre-sleep, post-sleep conditions?*	*How does the intervention influence the user's sleep patterns, perceptions, behaviours, or sleep environment over time?*	
			Implications for Others *How does the intervention affect people who share the space? What changes might they experience?*	
Artifact	**Data**	**Interaction**	**Sonic Approach**	**Playback Mode**
What is the tangible, physical or digital thing being created?	*What information, signals, or content drives the experience? Is this produced real-time or static?*	*How does the user engage and interact with it? (Passive/ active control, physical interaction, etc.)*	*What kind of sound design will the intervention use? (Functional e.g. alarms, notifications, data-driven e.g. sonification, soundscapes, music, narrative, digital sonic interactions, mechanical sonic interactions, etc.)*	*How will it be delivered? (e.g., headphones, speakers in space, mobile device, AR headset.)*
				Listening *What kind of listening do you expect to elicit? Active, attentive, passive, background, reflective...*

Intervention: Defining what you are creating and how it works?

7.6 Examples of post-its from workshop 1

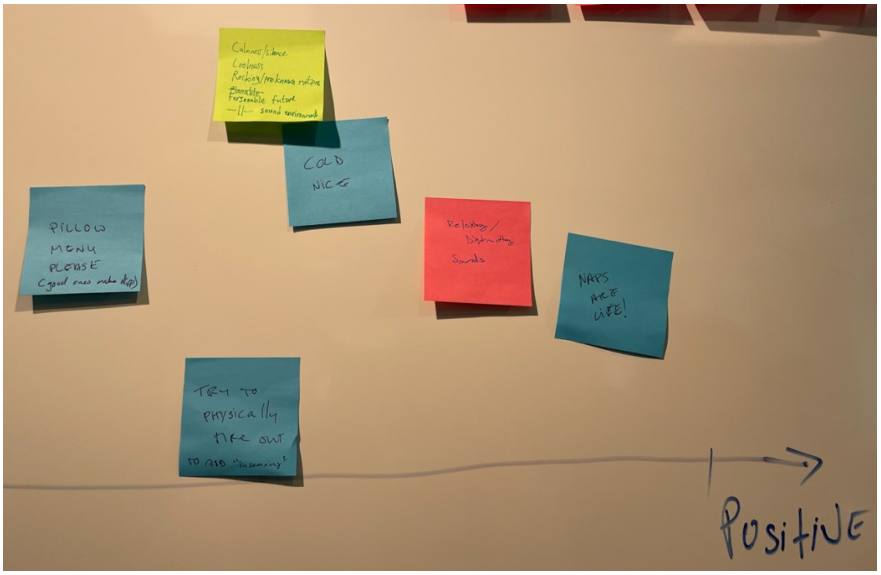


Figure 15 Post-its reflecting positive contextual factors.

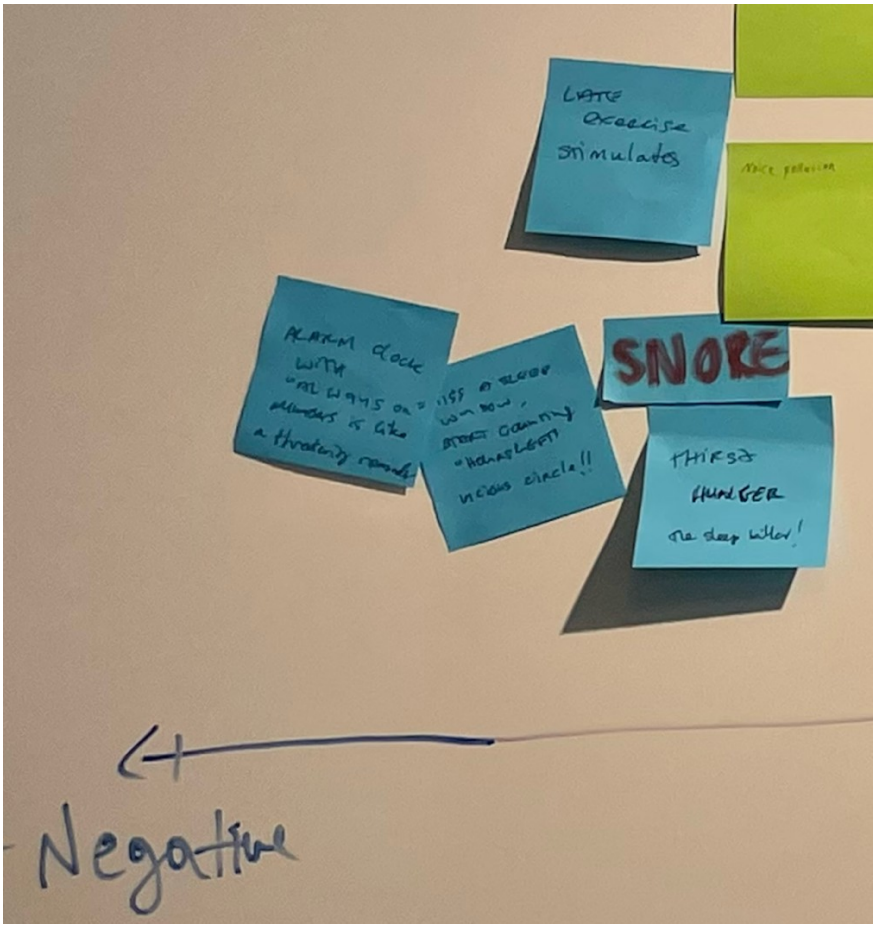


Figure 16 Post-its reflecting negative contextual factors.

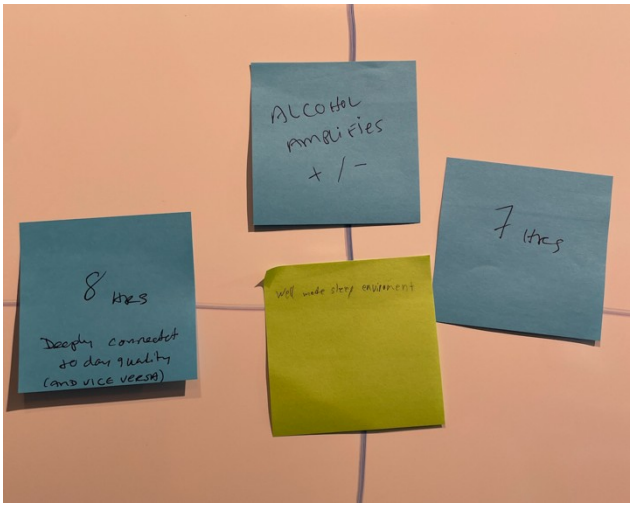


Figure 17 Post-its in the middle of the contextual factors space.

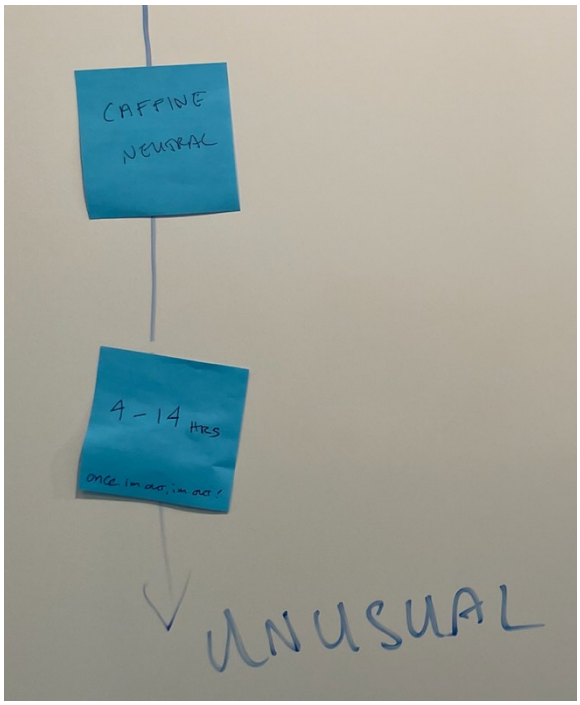




Figure 18 Post-its reflecting unusual contextual factors and the common contextual factors.

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